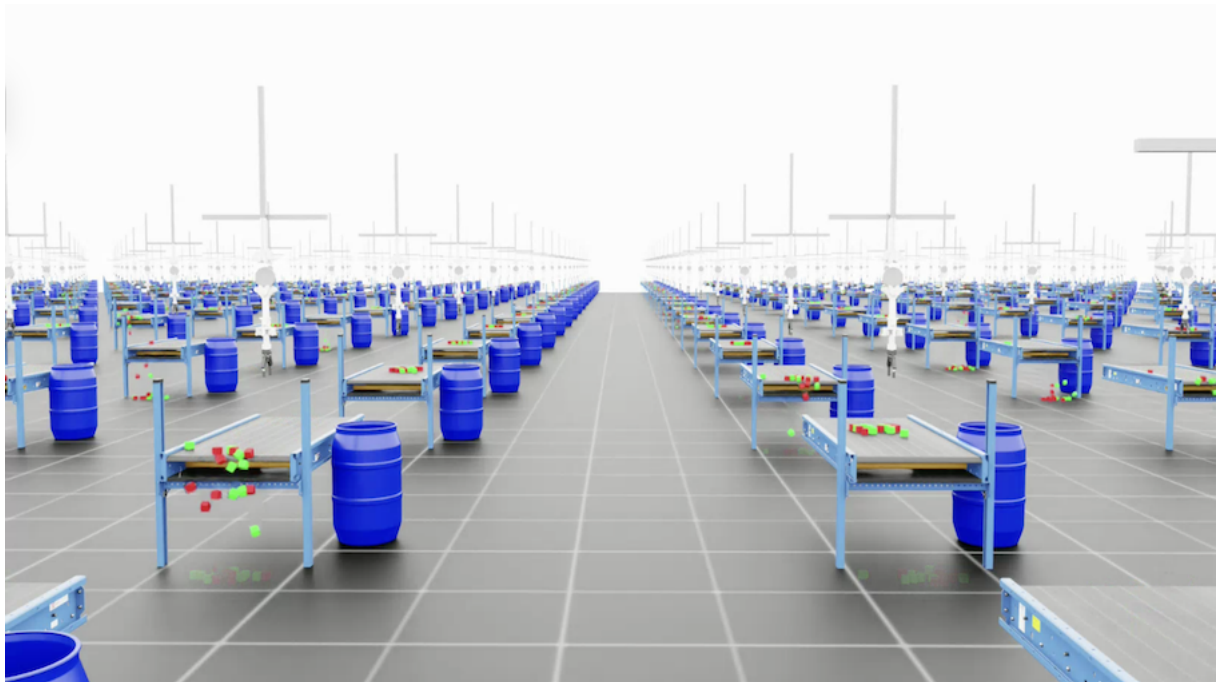


Simulative Construction and Verification of a Tendon-Driven Robot for Conveyor-Based Pick-and-Place Tasks Using Isaac Lab

Project Thesis im Studiengang Computational Engineering M.
Sc.

Friedrich-Alexander-Universität Erlangen-Nürnberg
Lehrstuhl für Fertigungsautomatisierung und Produktionssystematik
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Erlangen, den 9. März 2026

Georg Meyer

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List of Abbreviations

API	Application Programming Interface
CAD	Computer-Aided Design
CPU	Central Processing Unit
DoF	Degree of Freedom
DR	Domain Randomization
dVRK	da Vinci Research Kit
FEM	Finite Element Method
FK	Forward Kinematics
GAE	Generalised Advantage Estimation
GPU	Graphics Processing Unit
IK	Inverse Kinematics
IL	Imitation Learning
MDP	Markov Decision Process
MJCF	MuJoCo XML Format
MPC	Model Predictive Control
MPM	Material Point Method
PBD	Position-Based Dynamics
PD	Proportional-Derivative
PGS	Projected Gauss–Seidel
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RGB-D	Red-Green-Blue plus Depth
RL	Reinforcement Learning
ROS	Robot Operating System
SDF	Signed Distance Field
TGS	Temporal Gauss–Seidel
URDF	Unified Robot Description Format
USD	Universal Scene Description
VBD	Vertex Block Descent
XML	Extensible Markup Language
XPBD	Extended Position-Based Dynamics

List of Symbols

$\mathcal{A}$	Action space
A_t	Advantage estimator
ϵ	Clipping parameter (PPO)
γ	Discount factor
h	Simulation time step
$J(\theta)$	Expected discounted return
k_j	Stiffness parameter (PBD)
$\mathbf{M}$	Mass matrix
π_θ	Policy (parameterized by θ)
$r(s_t, a_t)$	Reward function
$\mathcal{S}$	State space
θ	Policy parameters

0 Guideline

0.1 Format and Style

0.1.1 Page Layout and Typography

- Paper format: white DIN A4 paper
- Font: Arial, 12pt (in this LaTeX template, the sans-serif option achieves this)
- Line spacing: 1.2 (the template uses 1.25)
- Text alignment: justified (Blocksatz) with automatic hyphenation enabled
- Font emphasis: use only in exceptional cases; headings follow the template style and must not be underlined
- Bullet points: only square bullet points (FAPS green), never round ones; use the template's built-in list style
- FAPS color scheme: FAPS-Grün (RGB 151, 193, 57) and white

0.1.2 Page Numbering and Binding

- Page numbering starts with the introduction (Arabic numerals)
- Directories (table of contents, list of figures, etc.) receive separate Roman numeral numbering; the cover page is not counted
- Print single-sided unless otherwise agreed with the supervisor
- Two bound copies required: adhesive binding with clear cover foil and green (FAPS green) back sheet
- Each copy must include a CD/DVD containing:
 - Digital version of the thesis (PDF)
 - All digital sources (papers, scanned documents, websites)
 - All images, tables, measurement data, models, and other data

0.1.3 Section and Chapter Formatting

- Chapters are numbered numerically with at most three levels (e.g., 1.1.1); no trailing dot after the last digit

-
- If a subsection X.1 exists, there must also be at least an X.2 (applies to all levels)
 - Main chapters should have at least 3 subchapters; aim for a balanced structure (e.g., 5 chapters at level 1, each with 4 at level 2, each with 3 at level 3)
 - Headings in the table of contents must match the headings in the text verbatim
 - Every heading must be followed by actual content (including introductory text at the start of main chapters)

0.1.4 Paragraphs, Spacing, and Whitespace

- Each paragraph must consist of at least two sentences
- No blank line between paragraphs within a chapter; one blank line before the next heading at the end of a chapter
- No double spaces anywhere in the document; use search-and-replace to eliminate them before submission
- Use non-breaking spaces (in LaTeX: ~) between:
 - Titles and names: “Prof. Dr. Beispielhausen”
 - Date expressions: “19. September 2007”
 - Abbreviations: “z. B.”, “u. a.”, “d. h.”
 - Numbers and units: “44 mm”
 - Numbers and words: “1.350 Personen”, “Nr. 5”
 - Citation references
- Check automatic hyphenation carefully for errors (e.g., incorrect compound-word breaks)

0.1.5 Figures

- Every figure receives a caption below it (Unterschrift), formulated so the figure is understandable without the surrounding text
- All figures are numbered consecutively with Arabic numerals and listed in the list of figures
- Every figure must be referenced in the body text
- Title image: no caption, no numbering, not included in the list of figures
- Appendix figures: numbered within each appendix separately (e.g., Appendix A – Figure 1); not included in the list of figures
- Ensure sufficient image quality; recreate figures if originals are blurry, low-resolution, or scanned
- Use the same font in figures as in body text (Arial); font size 10pt is acceptable but must be consistent across all figures

-
- All figure components must be fully labeled: photo components, axes, parameters, legends with complete process parameters
 - Fill areas with color only if the color conveys meaningful information; otherwise keep figures clean and minimal
 - For recreated/modified figures from sources, cite with “in Anlehnung an [xy]” (based on [xy])
 - Use a consistent color scheme across all figures; preferably FAPS colors unless industry cooperation requires otherwise
 - Use a self-created graphic on the first page of the thesis

0.1.6 Tables

- Every table receives a caption above it (Überschrift) with a meaningful, descriptive title
- Tables are numbered consecutively with Arabic numerals and listed in the table directory
- Every table must be referenced in the body text
- Use the same font as in body text; maintain consistent font size across all tables
- Preferred alignment inside tables: left-aligned or centered (not justified)
- If a table spans a page break, do not break within a row; repeat the table header on the new page
- Extensive tables belong in the appendix

0.1.7 Diagrams

- Follow DIN 461 for axis labeling in diagrams
- Label axes with an arrow in the direction of increase (except for trial numbers etc.)
- Replace the axis label at the second-to-last tick mark with the unit
- Display all relevant information within the diagram
- Place a frame around diagrams
- Use Arial font inside diagrams with a legible font size
- Diagrams are captioned and numbered as figures (no separate numbering type)

0.1.8 Formulas

- Formula numbers are enclosed in round parentheses

-
- Numbering is chapter-based in the form X.Y (X = chapter number, Y = formula number within the chapter)
 - Numbering restarts within each main chapter
 - Variables used in a formula must be explained directly below the formula in tabular form

0.1.9 Appendix

- Technical drawings, extensive tables, supplementary documents, etc. go into appendices
- Each appendix starts on a new page and may span multiple pages
- Appendices are numbered with consecutive uppercase Latin letters (Appendix A, Appendix B, etc.)

0.1.10 Eidesstattliche Erklärung (Statutory Declaration)

- A signed declaration of independent authorship must be included on a separate page without a page number, placed after the cover page
- It must be signed by hand

0.2 Thesis Structure and Content

0.2.1 Required Document Order

1. Deckblatt (Cover Page)
2. Erklärung des Kandidaten (Statutory Declaration)
3. Inhaltsverzeichnis (Table of Contents)
4. Abbildungsverzeichnis (List of Figures)
5. Tabellenverzeichnis (List of Tables)
6. Abkürzungsverzeichnis (List of Abbreviations)
7. Einleitung und Zielstellung (Introduction and Objective)
8. Grundlagen / Stand der Technik (Background / State of the Art)

-
9. Handlungsbedarf (Research Gap / Need for Action)
 10. Methodenteil (Methodology)
 11. Ergebnisteil (Results)
 12. Diskussionsteil (Discussion)
 13. Zusammenfassung und Ausblick (Summary and Outlook)
 14. Literaturverzeichnis (Bibliography)
 15. Anhang (Appendix)
 16. Lebenslauf (Curriculum Vitae)

0.2.2 Page Count Guidelines

- Bachelorarbeit: ca. 60–80 pages
- Projekt-/Studienarbeit: ca. 60–80 pages
- Master-/Diplomarbeit: ca. 80–120 pages
- These counts exclude appendices and directories

0.2.3 Red Thread (Roter Faden)

- A clear red thread must run through the entire thesis, starting from the central research question in the introduction
- The reader must always be able to see how any part of the thesis relates to the research question
- The thesis must proceed purposefully and not lose itself in tangential topics
- All claims and statements must be causally justified; results must not merely be listed but convincingly explained
- Each main chapter must begin with a short introduction motivating the content and end with a transition to the next
- Verify the red thread by checking all headings alone (without reading the text) — the logical flow should be recognizable

0.2.4 Chapter: Einleitung und Zielstellung (Introduction)

- Length: approximately 2 pages, maximum 3 pages

-
- Purpose: establish contact with the reader; provide orientation (“What can the reader expect?”)
 - Content requirements:
 - Introduce the topic and its general background; explain why it is relevant
 - Present the specific research question
 - Justify the scope and boundaries of the thesis (what is included, what is excluded, and why)
 - Optionally describe the state of research (to show what exists and where gaps are)
 - Explain the methodological approach
 - Outline the structure of the thesis for the reader

0.2.5 Chapter: Grundlagen und Stand der Technik (Background and State of the Art)

- Clarify central concepts and define relevant terms
- Present the current state of the art objectively and without judgment (evaluation happens in the next chapter)
- Only cover theories, definitions, and knowledge directly relevant to the research question; avoid overly broad coverage
- If multiple competing theories exist, provide an overview
- Definitions are only needed for terms not commonly known by domain experts
- Present content in your own words; near-verbatim copying or translation from sources counts as insufficient
- Use primarily English-language sources where possible
- Prefer journal articles and conference papers over websites
- Avoid popular-science sources (e.g., Schülerduden, Brockhaus); only quote newspapers for current events
- Internet sources are acceptable only if no current literature exists elsewhere (e.g., software documentation); verify their credibility

0.2.6 Chapter: Handlungsbedarf (Research Gap)

- Critically discuss the literature from the previous chapter regarding unsolved problems and errors
- Derive current deficits and research gaps; formulate the specific research question
- Explain who considers the problem significant and what its broader impact is

-
- Describe the contribution the thesis makes to both research (theory, models, methods, facts) and practice
 - Narrow the problem statement into a concrete, achievable objective with clear expected outcomes

0.2.7 Chapter: Methodenteil (Methodology)

- Together with Results and Discussion, this forms the main body (2/3 of the thesis)
- Describe the investigation plan, procedures, and all steps and tools in sufficient detail for reproducibility
- Justify why specific methods were chosen over alternatives (pros/cons analysis)
- For experimental studies: describe the population and sample in all relevant characteristics; justify sample adequacy
- Scientific criteria for methods: objectivity, reliability, and validity
- The description must enable other researchers to reproduce the investigation
- Only describe analysis procedures in detail if they are novel or non-standard

0.2.8 Chapter: Ergebnisteil (Results)

- Present results of the investigation objectively and without interpretation
- Structure the chapter logically, aligned with the theory and methodology sections
- Use figures, tables, and diagrams to visualize results
- No interpretation or evaluation at this stage

0.2.9 Chapter: Diskussionsteil (Discussion)

- Summarize and critically discuss the results
- Evaluate and interpret results in relation to the original problem and objective from the introduction
- Argue the significance for scientific discourse and practical implications

0.2.10 Chapter: Zusammenfassung und Ausblick (Summary and Outlook)

- Maximum 3 pages
- Reconnect to the central problem and objective from the introduction
- Compactly summarize the most important results

-
- Place results in a broader context
 - Address:
 - What are the key findings with respect to the research questions?
 - How can the results be interpreted or evaluated?
 - What questions remain open, and why?
 - What do the results mean for future work on this topic?
 - Where are opportunities for further research?
 - Discuss the limitations of the investigation
 - Provide suggestions for future work and improvements based on newly emerged questions

0.3 Scientific Writing Style

0.3.1 General Principles

- Write in present tense (Präsens) and nominal style (Nominalstil)
- Use impersonal form throughout; never use “ich” (I) or “man” (one/you)
- Develop the red thread of argumentation before formulating sentences
- Every sentence and every claim must be precise and well-placed
- Write result-oriented: “Die Einflüsse führen zu Veränderungen.” not “Im Rahmen der Arbeit wurden untersucht...”

0.3.2 Sentence Construction

- Use passive voice sparingly — prefer active formulations
- Avoid sentences with “man” entirely
- Be careful with infinitive constructions
- Avoid chaining too many nouns together (“Substantivaneinanderreihungsproblem”)
- No nested or overly complex sentences (Schachtelsätze)

0.3.3 Grammar

- Avoid relative time references (“kürzlich”, “später”, “modern”) since the thesis should remain valid for years
- Do not use subjunctive mood (könnte, sollte, müsste, dürfte)

0.3.4 Technical Language

- Use precise adjectives; avoid vague terms like “innovativ” or “intelligent”
- No pleonasms (redundant word pairs)
- Do not artificially inflate terms
- Be cautious with superlatives and comparative forms
- Do not invent words
- Use anglicisms and foreign words sparingly; only when no equivalent German term exists
- Use precise technical terminology; avoid colloquial language
- Use technical terms consistently throughout (e.g., always “CAD-Modell”, never alternating with “CAD Modell”)

0.3.5 Expression and Phrasing

- Spell out numbers from zero to twelve; use digits from 13 onward
- Avoid colloquialisms and journalistic clichés
- Do not use rhetorical questions; make clear statements instead
- Avoid vague qualifiers: “im Allgemeinen”, “in der Regel”, “beispielsweise”, “teilweise”
- Avoid filler words: “nämlich”, “also”, “so”, “eben”
- Avoid repetition of both words and arguments
- Always formulate concretely and precisely; define unambiguously; enumerate completely; avoid generalizations
- Use nominal style: instead of “wird bearbeitet”, write “Zur bedarfsgerechten Bearbeitung ist auf die Einhaltung von xy zu achten.”

0.3.6 Content Guidelines

- Do not document iterative development processes or suboptimal intermediate solutions; present only the final approach
- Do not describe device/hardware development in excessive detail; focus on scientifically relevant, generalizable, transferable aspects
- Use direct quotes rarely and only pointedly; prefer evaluating and developing ideas further in your own words
- Introduce abbreviations only when necessary for space; spell out on first use and include in the abbreviation index

-
- Emphasize scientific quality rather than engineering challenges
 - Emphasize fundamental, generalizable, and transferable findings rather than application-specific details
 - Instead of giving examples, enumerate completely, structure, and evaluate
 - Make all enumerations and representations as complete and non-overlapping as possible (MECE principle)
 - Always support claims with your own analyses or cited sources
 - No implied personal opinion (e.g., “jedoch”, “überaus”) and no subjective assessments except in the Outlook section
 - Where possible, summarize each chapter’s key message in a figure for quick reference

0.3.7 Introductions and Transitions Between Sections

- Every main chapter or main section must begin with a brief introduction and end with an outlook/transition to the next section
- Example introduction: “Hasen und Igel sind, betrachtet man ihre Laufgeschwindigkeit, auf einem vollkommen unterschiedlichen Niveau.”
- Example transition: “Deshalb wird im folgenden Abschnitt der Wettlauf zwischen Hase und Igel näher betrachtet.”
- When introducing detailed explanations, name all relevant points in the introductory sentence
- Bad: “Im Folgenden werden die beiden Protagonisten näher betrachtet.”
- Good: “Im Folgenden werden die beiden Protagonisten Hase und Igel näher betrachtet.”

0.3.8 Consistency and Formatting Rules

- Introduce each abbreviation exactly once and then use it consistently
- Never write a paragraph consisting of only one sentence (minimum two sentences)
- Never write a list with only one bullet point (minimum two)
- Use nominal style for bullet-point items
- Maintain consistent spelling of expressions throughout (e.g., always “Aerosol-Jet-Druck”, never “Aerosoljet-Druck”)

0.4 Citations and Bibliography

0.4.1 Citation Style

- Citation style: DIN ISO 690 (FAPS custom style via `bibstyleFAPS.bst`)
- Reference sources in the text using numbers in square brackets: [xy]
- Page numbers are not required in citations
- Multiple consecutive citations use separate brackets: [xy] [yz], not [xy; yz]
- The bibliography is ordered chronologically and numbered consecutively; no grouping by source type
- Only actually cited works appear in the bibliography

0.4.2 Citation Placement

- A citation before the period refers to the specific sentence [xy].
- A citation after the period of the last sentence refers to the entire paragraph. [xy]

0.4.3 Direct Quotes

- Must be letter-perfect and character-perfect reproductions
- Enclose in quotation marks; cite the source at the end: “...” [20]
- Quote foreign-language texts in the original language; do not translate
- Mark own additions or rearrangements with square brackets
- Mark omissions with [...]
- Quotes longer than three lines: indent as a block, set in italic, omit quotation marks
- Do not correct errors in the original; mark them with [!] or [sic!]
- Emphasis within quotes: append [Hervorhebung durch Verfasser]

0.4.4 Indirect Quotes

- Paraphrased ideas from other sources; not placed in quotation marks
- Still require a source reference in the same placement rules as above

0.4.5 Secondary Citations

- Always prefer citing the original source

-
- Use secondary citations only when the original is unavailable; reference the secondary source in the bibliography

0.4.6 Plagiarism

- Provide a source reference for every piece of knowledge that goes beyond common knowledge
- Originality and independence are the most important quality criteria
- Even when rephrasing or summarizing, the source must be cited
- It must be unambiguously clear to the reader at all times what is borrowed intellectual property

0.4.7 Source Quality Requirements

- Every source must be: (1) citeable (publicly accessible), (2) citeworthy (scientific quality), and (3) relevant
- Prefer English-language sources
- Prefer journal articles and conference papers over websites
- Avoid websites as sources wherever possible; especially avoid Wikipedia, hausarbeiten.de, and similar
- If no conventional sources exist (e.g., for software topics), internet sources are permissible with critical evaluation
- Prioritize current sources from renowned journals or scientists over older work
- Quality indicators: scientific question, clear structure, precise terminology, theoretical foundation, appropriate methodology

0.4.8 Electronic Sources

- Books, journals, and publications that also exist in print are not “electronic sources” even when accessed online
- Each source entry must answer: Who? What? Where? When?
- Use DOI when available; otherwise provide the full URL and access date
- Do not insert hyphens in URLs (could be misinterpreted as part of the address)
- If author is unknown: use “o. V.” (ohne Verfasser); if date is unknown: use “o. J.” or “o. D.”
- Schema: Nachname, Vorname (Jahr): Titel. [ggf. weitere Angaben]. DOI oder online unter: URL (Abrufdatum). No period after DOI.

0.5 Literature Research

- Recommended databases: OPAC (FAU library), Google Scholar, IEEE Digital Library, Scopus
- Strategy: create a search table with key terms, synonyms (abstract and specific) across multiple columns
- Use primarily English-language sources where possible
- Prefer journal articles and conference papers; avoid websites
- Submit a detailed outline and relevant literature to the supervisor within 2 weeks (for Abschlussarbeiten)

0.6 Figures and Graphics — Additional Rules

- Use a self-created graphic on the first page of each chapter or major section
- If independent creation is not possible or the graphic is assembled from multiple sources, provide a source note on the first page where it appears
- Poorly formatted, inconsistent, blurry, or illegible graphics create a bad impression; avoid copies and scans
- Recreate relevant graphics, diagrams, formulas, and tables to ensure uniform formatting and meet formal requirements
- Maintain a balanced ratio of text to graphics (e.g., avoid placing two graphics directly after each other)
- Ensure consistent design across all graphics: shapes, fonts/sizes, colors

0.7 Review Checklist Before Submission

0.7.1 Content Review

- Verify all headings form a coherent red thread (recognizable without reading the text)
- Verify transitions and introductions exist at the beginning of every main chapter
- Is the central research question clearly developed in the introduction?
- Is the relevant literature reviewed with appropriate focus (not quantity, but quality of selection)?
- Can the derivation of the target concept from the analysis be followed?
- Are strengths and development potentials considered in the solution space?

-
- Is the proposed solution feasible and implementable?
 - Does the outlook present innovative approaches for future work?
 - Are all tables and figures clear and understandable?
 - Is the context of existing literature properly established?
 - Is there a concise summary that still covers all essential points?
 - Is the overall structure clear and comprehensible?
 - Is the scope sufficient yet as concise as possible?
 - Were all key work items of the thesis topic fulfilled and the objectives achieved?

0.7.2 Language and Expression Review

- Check for scientific, objective tone throughout
- Check for linguistic expression quality
- Run spell-check and grammar-check
- Have at least one other person proofread the thesis

0.7.3 Visual and Formatting Review

- Check in single-page view for equal spacing between sections
- Verify balanced ratio of text and graphics (no dense clustering of figures)
- Verify consistent design of all graphics (shapes, fonts, sizes, colors)
- Eliminate all double spaces
- Check for sensible page breaks (no orphan headings, no broken table rows)

0.8 Grading Criteria

- Systematic approach and methodology
- Personal effort and creativity
- Fulfillment of the thesis assignment
- Independence during the work
- Application of domain expertise
- Discussion and critical evaluation of results
- Presentation of the initial situation
- Thoroughness, completeness, and comprehensibility of the presentation
- External form and appearance of the thesis
- Expression and style

-
- Spelling and grammar
 - Performance is measured as work divided by time — high quality within deadlines demonstrates strong performance

0.9 Presentation Guidelines (BA/PA)

- Duration: 20 minutes $\pm$ 10% deviation
- Bring your own laptop; have a backup on a USB drive
- Submit slides to the supervisor several days before the presentation date for review
- Follow FAPS presentation guidelines
- Dress code: at minimum business casual
- Structure:
 1. Title slide
 2. Motivation slide (between title and table of contents)
 3. Table of contents slide (all relevant topics except bibliography)
 4. Content slides
 5. Summary slide
 6. Closing slide
 7. Bibliography (hidden; only shown if asked during Q&A). Cite sources on individual slides in (Author, Year) format

1 Introduction

1.1 Motivation and Context

- Introduce the industrial relevance of autonomous robotic manipulation for textile recycling and waste sorting on conveyors
- Highlight the challenges of using tendon-driven robots: compliant design, complex actuation, need for simulation-based development
- Emphasize the role of simulation (NVIDIA Isaac Sim) and reinforcement learning as an enabling approach for developing control policies without physical hardware iteration
- Frame the project thesis as the foundational simulation and validation stage of a larger multi-agent textile sorting project

1.2 Problem Statement and Objectives

- State the thesis problem: constructing, validating, and controlling a tendon-driven robot in simulation for conveyor-based pick-and-place tasks
- List the research objectives:
 1. High-fidelity simulation environment (conveyor, drum, objects) in Isaac Sim 5.1.0
 2. Robot model integration and validation (5-DOF tendon-driven arm + base + gripper)
 3. RL-based reach and cube place tasks with PD and tendon-driven variants
 4. Tendon wrist transfer experiment on a standard industrial robot arm
- Scope delimitation: this thesis does NOT address cloth simulation, sorting (selectivity), multi-agent coordination, or sim-to-real transfer; these are explicitly deferred to the subsequent master thesis

1.3 Thesis Outline

- Brief summary of chapters 2–7 for reader orientation

2 Background

2.1 Reinforcement Learning for Robotic Manipulation

- MDP formalism: states, actions, transitions, rewards, discount factor
- Policy gradient methods relevant to continuous control: PPO (primary), SAC, DDPG (for comparison context)
- On-policy (PPO) vs. off-policy approaches: stability, sample efficiency, suitability for GPU-parallel simulation
- Policy network architectures: MLP actor-critic, observation normalization
- The skrl library as the RL framework used in this thesis
- Prior successes of RL in simulation for manipulation (OpenAI Dexterous Hand, QT-Opt, DexMV)

2.2 Physics Simulation in Isaac Sim 5.1

- NVIDIA PhysX 5 as core solver, GPU-accelerated simulation
- Rigid body dynamics: TGS solver, articulated bodies, joints, contacts, friction
- Simulation parameters: time step, solver iterations, contact offsets
- Collision detection: mesh vs. convex decomposition vs. primitive shapes
- Brief mention of deformable simulation capabilities (PBD particle cloth, VBD) as context for future work — not used in this thesis

2.3 NVIDIA Isaac Sim and IsaacLab Architecture

- Isaac Sim 5.1.0: Omniverse platform, PhysX 5, RTX rendering, ROS 2 support
- IsaacLab: unified robotics learning framework, manager-based environment structure
- MDP workflow: observation managers, action managers, reward managers, event managers
- Comparison with alternative platforms (MuJoCo, PyBullet) to justify choice

2.4 Tendon-Driven Robot Kinematics and Modeling

- Tendon-driven actuation: cables generate joint torques, advantages for compliant manipulation
- The specific robot: 2-DOF linear base + 3-DOF tendon arm (Klein 2023) + Robotiq 2F-140 gripper
- Jacobian-transpose torque mapping: 5 cable tensions to 3 joint torques
- Challenges of simulating tendon-driven systems in Isaac Sim; decision to use PD drives instead of PhysX tendon features
- Workspace analysis fundamentals: FK-based Monte Carlo sampling, Yoshikawa manipulability index, inverse condition number, reachability density

2.5 Policy Learning Pipeline

- Observation space design for robot control tasks
- Action space: joint position targets, tendon tension targets
- Reward function design: dense shaping rewards, curriculum learning
- Episode termination and reset strategies
- GPU-parallel vectorized environments for training throughput

2.6 Prior Work

- TODO: Related work on tendon-driven robot simulation and RL-based pick-and-place

2.7 Robot and Scene Description Formats

- USD: extensible 3D scene graph for Isaac Sim
- URDF: robot kinematic/dynamic description
- URDF-to-USD import pipeline and modifications for the tendon arm
- Isaac Sim Robot Assembler for modular multi-component robots

3 Research Gap

3.1 Critical Analysis of State of the Art

- Tendon-driven robots are studied in control theory but rarely integrated with modern RL simulation frameworks
- Most RL manipulation work uses standard rigid-joint robots (UR, Franka); tendon-driven arms are underrepresented
- No existing work combines a tendon-driven arm with IsaacLab’s GPU-parallel RL training pipeline
- The transferability of tendon wrist mechanisms to standard arms is unexplored

3.2 Research Gap and Justification

- Gap: validated simulation of a tendon-driven robot in Isaac Sim with RL-based control for manipulation tasks does not exist in the literature
- Scientific contribution: methodology for integrating tendon-driven actuation in a modern GPU-parallel RL framework; comparative validation against Klein (2023)
- Practical contribution: a modular, reusable simulation environment for tendon-robot research; wrist transferability results

3.3 Refined Problem Definition

- Research question: How can a custom tendon-driven manipulator be faithfully modeled, validated, and controlled via RL in Isaac Sim for conveyor-based pick-and-place tasks, and does the tendon wrist mechanism transfer beneficially to a standard robot arm?
- Scope: simulation only; no physical hardware; no cloth; no sorting selectivity
- Expected contributions: validated robot model, RL pipeline, wrist transfer comparison

4 Methodology

4.1 Overview of Approach

- High-level system diagram: Isaac Sim scene → IsaacLab environment → skrl PPO → learned policy
- Development stages: (1) environment and robot setup, (2) validation, (3) RL tasks, (4) wrist transfer experiment

4.2 Simulation Environment Setup

- USD scene: ground plane, conveyor belt, plastic drum, dome lighting
- Object placement and randomization (initial positions, colours for curriculum)
- Physics parameters: time step, solver iterations
- Parallel environment configuration for GPU training
- Figure: annotated simulation scene

4.3 Robot Model Integration

- 2-DOF articulated base: USD construction, prismatic joints, travel ranges
- 3-DOF tendon-driven arm: URDF import from Klein (2023), modifications, revolute joints with PD drives
- Tendon actuation modeling decision: why PhysX tendon features were not used; PD drive approximation
- Robotiq 2F-140 gripper: asset library integration, specifications
- Modular assembly via Robot Assembler: two-step procedure (base+arm, then arm+gripper)
- Collision geometry: convex hull generation, self-collision verification
- Joint limits table (Table X)
- Figure: exploded view of assembled robot

4.4 Control Gain Tuning and Validation

- Isaac Sim Gain Tuner: methodology for automated stiffness/damping tuning
- Step-response testing: sinusoidal and step commands, metrics (settling time, overshoot, steady-state error)
- Comparison with Klein (2023) PID controller parameters
- Gravity compensation for ceiling-mounted configuration
- Figure: Gain Tuner output (commanded vs. actual trajectories)

4.5 Workspace Analysis

- Monte Carlo FK sampling: 200,000 samples, 4096 parallel environments
- Data collected: end-effector position, Jacobian, Yoshikawa manipulability, inverse condition number
- Voxelization and density aggregation
- Comparison with Klein (2023) task geometry
- Visualization: 3D scatter plots, 2D heatmap cross-sections

4.6 Reinforcement Learning Setup

- **Reach Task:** observation space (joint states, target pose), action space (joint position targets), reward (pose distance + orientation), PD-driven and tendon-driven variants
- **Cube Place Task:** observation space (joint states, cube pose, drum pose, grasp status), action space, reward (approach + grasp + lift + place), curriculum (green-only $\rightarrow$ green+red)
- PPO hyperparameters (learning rate, discount, network architecture, number of environments)
- Reproducibility table: software versions, hardware

4.7 Tendon Wrist Transfer Experiment

- Experimental design: UR10 with native rigid wrist vs. UR10 with tendon-driven wrist replacement
- Control variable: wrist type; all other parameters (RL config, task, scene) identical

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- Tasks: reach accuracy, place success rate
 - Metrics: final pose error, success rate, learning speed
 - Integration procedure: wrist USD asset swapped via Robot Assembler

5 Results

5.1 Robot Simulation Validation Results

- Step-response plots per joint: commanded vs. actual position
- Quantitative metrics: final stiffness/damping, settling time, overshoot, steady-state error
- Comparison with Klein (2023) PID results; discussion of differences
- Collision geometry validation across full joint range

5.2 Workspace Analysis Results

- 3D reachable workspace envelope and volume estimate
- 2D cross-section heatmaps by reachability density and manipulability
- Yoshikawa manipulability and inverse condition number distributions
- Coverage of desired task workspace; practical implications for object placement
- Comparison with Klein (2023) workspace (without 2-DOF base)

5.3 Reach Task Results

- Learning curves (reward vs. training steps) for PD and tendon-driven variants
- Final pose accuracy: position error, orientation error
- PD vs. tendon comparison: learning speed, final accuracy
- Computational cost: wall-clock time, GPU memory, environment steps

5.4 Cube Place Task Results

- Learning curves with curriculum progression
- Success rate: cube in drum, colour selectivity (green placed, red ignored)
- PD vs. tendon comparison
- Qualitative policy behavior description with keyframe figures

5.5 Tendon Wrist Transfer Results

- UR10 native wrist vs. UR10 + tendon wrist: reach accuracy comparison
- Place success rate comparison
- Learning curves comparison
- Statistical significance of observed differences

6 Discussion

6.1 Evaluation of Outcomes

- To what extent were the thesis objectives achieved?
- Robot model validated against Klein (2023); deviations explained by simplified tendon model
- RL pipeline functional for both reach and place; tendon-driven variant achievable but with different convergence characteristics

6.2 Tendon Wrist Transferability

- Interpretation of wrist transfer results
- Under what conditions does the tendon wrist improve or degrade performance?
- Implications for robot design: when is a compliant wrist advantageous?

6.3 Limitations

- Simplified tendon model (PD drives, no cable stretch or backlash)
- Rigid objects only; no deformable cloth tested
- Simulation-only results; no sim-to-real validation
- Wrist experiment limited to one standard arm type

6.4 Practical Implications

- Modular simulation pipeline reusable for future tasks
- Workspace analysis informs physical task design
- Wrist results guide hardware decisions for the master thesis

7 Conclusion

7.1 Conclusion

- Recap accomplishments: validated simulation, RL-controlled tendon robot, wrist transferability assessment
- Answer the research question concisely
- Highlight key quantitative results

7.2 Outlook

- Direct next steps (master thesis scope):
 1. Conveyor-based sorting with selectivity (T-shirt vs. non-target)
 2. Deformable cloth simulation integration
 3. Second robot arm for stretching and inspection
 4. Camera-based damage classification
 5. Multi-agent coordination
 6. ROS 2 integration for sim-to-real transfer
- More realistic tendon model (cable dynamics, backlash)
- Domain randomization for transfer robustness